

Programmable Sovereignty: Lean-Attestable Constitutions Over Capability-Bounded Federated Receipts

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Abstract

Agent systems already allocate authority, settle costs, and emit records that become accounting and dispute evidence, yet their governance reduces to prompts, API keys, dashboards, and after-the-fact logs. We give a substrate in which a polity is a triple (T, C, K) : a closed receipt-admission scope T , a Merkle-rooted citizenship roster C , and a constitution K expressed as a finite list of predicates evaluated by a runtime kernel and discharged by machine-checked proofs over a bounded model. Admission, capability attenuation, treaty intersection, and constitutional amendment become decidable operations on this triple, and each emits a canonically signed receipt only when K and T accept. We implement the construction in Rust with companion proofs in Lean 4, prove theorems for treaty predicate intersection, ladder-floor stability, and amendment refinement, and demonstrate the construction with a three-vendor buyer-closure backed by replay fixtures. Treaty intersection composes with ladder-floor reduction without changing admission; amendment enactment is conditioned at the type level on a refinement witness and is unconstructable in the well-typed runtime path without one; the three-vendor demonstration replays both admitted and denied paths in the same canonical schema. The paper contributes the formal model, the substrate implementation, and an evaluation across dispatch admission, treaty intersection, selective disclosure, and replay-corpus stability.

1 Introduction

Tool-using agents already behave like institutional actors. They receive delegated authority, buy services, call tools hosted by other organizations, emit records that become accounting evidence, and trigger irreversible actions. Deployments can record who configured a prompt or approved an integration; they cannot prove that a later cross-vendor action satisfied a declared constitution, a treaty, and a capability boundary before it crossed the trust boundary. We close that gap by reducing institutional authority to four decidable operations on a polity triple (T, C, K) : admission against the receipt scope T , attenuation within the citizenship roster C , bilateral treaty intersection between two constitutions K_1 and K_2 , and amendment refinement within K . Each operation is checked by Rust at the kernel boundary, attested by a named Lean theorem, and signed into a canonical receipt admitted only when the constitution and the scope accept.

The construction rests on three substrate conditions. Authority is carried by attenuating capability tokens, not ambient credentials. Every kernel decision is recorded as a signed receipt over canonical JSON bytes. Membership, treaty, and amendment rules are predicates whose safety claims map to machine-checked proof artifacts. Chio implements that stack: an untrusted agent presents a capability, a trusted runtime kernel evaluates scope and guard

predicates, tool servers remain isolated, a capability authority issues and revokes bounded grants, and the receipt log records the decision.

The polity claim is declarative. A Chio polity has final authority over its own receipt-admission boundary; it has none over physical persons, land, or external courts. Its territory is a receipt namespace, its citizens are Decentralized Identifiers (DIDs) per the W3C DID Core specification [66] or kernels admitted into that namespace, its border is the runtime admission hook, and its courts are verdict artifacts over signed receipts. The substrate gives a constructive test for any further claim: it lives or fails as an admission predicate, treaty artifact, proof term, or replayable script. Section 7 positions this reading against the Montevideo Convention and the network-state literature; Section 9 states the legal scope.

The contributions are five falsifiable artifacts.

- A formal model relating polity scope, constitutional predicates, receipt admission, treaty intersection, and amendment refinement (Section 4).
- Two load-bearing Lean theorems and two definitional refinement bridges in `formal/lean4/Chio/Chio/Treaty/Intersection.lean`: `treaty_admission_iff_predicate_intersection` and `treaty_admission_stable_under_ladder_floor` (load-bearing), with `amendment_admissible_iff_backward_refinement` and `amendment_without_refinement_rejected` (definitional, anchoring the type-level enactment invariant) (Section 4).
- An implementation of the model in the Chio runtime, covering treaty scope and ladder intersection, pre-dispatch admission, strict bilateral DSSE verification, and a multi-lane anchor whose maturity varies by lane (Section 5).
- A three-vendor buyer-closure demonstration built on a fixture-backed generator at `examples/chiodos-3vendor/`, exercising every load-bearing predicate in the substrate (Sections 5 and 6).
- An evaluation that measures dispatch admission, treaty intersection, selective disclosure, and replay-corpus stability, and that names which lanes and primitives have no current measurement (Section 6).

2 Background

Capability systems replace ambient authority with explicit references that carry rights. KeyKOS and EROS demonstrated operating systems built around this discipline [29, 55]. Capsicum extended capability mode and rights-bearing descriptors into UNIX [68]. seL4 then connected a microkernel implementation to machine-checked proofs of functional correctness [34, 54]. Chio inherits the capability primitive and applies it to tool invocation rather than process-local file descriptors: authority is a signed object the kernel evaluates before dispatch.

Object-capability research sharpened the programming-language side of the same idea. Miller’s dissertation framed authority as graph reachability and attenuation [40]; Murray analyzed object-capability patterns with formal security properties [41]; Drossopoulou and Noble argued that capability policy needs explicit specification rather than folklore [23]. Fuchsia’s component framework supplies a production comparison: a component uses a capability only when the declared route exists [26]. Chio differs by signing the authorization decision into a portable evidence object that crosses organizational boundaries.

Transparency logs solve a different problem: public append-only evidence. Certificate Transparency logs TLS certificates so monitors can discover misissuance [35]. Sigstore’s Rekor generalizes signature transparency for supply-chain metadata [56, 57]. in-toto and SLSA bind build steps, subjects, and provenance claims to signed artifacts [58, 61]. Chio’s receipt is not artifact provenance; it is the decision record for a tool call that may spend money, mutate state, or cross a treaty boundary.

On-chain governance binds protocol change to votes recorded on a public chain. Compound Governor Bravo ties change to token-weighted proposals and timelocks [13]; the 2024 Proposal 289 attack passed a \$24M extraction proposal under low voter turnout before later rescission by negotiated settlement, demonstrating that token-weighted enactment without proof obligation is exploitable by capital concentration [14]. Polkadot OpenGov uses tracks and referenda for runtime and treasury decisions [46]. Helios shows that voters can audit election results without trusting a web server [1]. Kleros and Aragon Court place dispute resolution on-chain with incentives and jurors [3, 37]. Chio’s amendment path differs in kind: a vote may approve a proposed constitution, but enactment requires a Lean term witnessing backward refinement.

Legal-philosophical work on the rule of recognition gives the closest existing framework for what Chio’s admission predicate is structurally doing: Hart argued that every legal system reduces to a rule of recognition by which officials identify which rules count as law of the system, and modern positivism after Raz and Schauer has continued the program of making that rule explicit [30, 48, 51]. Chio’s admission predicate K is a machine-checkable instantiation of that rule of recognition over a receipt log; Section 7 returns to this framing.

Network-state claims have already been tested against real legal substrate. Próspera’s Honduran ZEDE project became a sovereignty dispute and a multibillion-dollar arbitration claim rather than a settled jurisdictional form [33, 49]. FTX’s Bahamas-based institutional theater did not substitute for verifiable custody or governance; Bankman-Fried was sentenced by a U.S. federal court on March 28, 2024 [64]. Tornado Cash showed the opposite edge: code and governance artifacts can be legally salient in ways that exceed normal compliance categories, and Treasury removed Tornado Cash from the SDN list on March 21, 2025 after the Fifth Circuit litigation [63, 65]. Chio reads these as failure modes for any state-flavored claim that lacks substrate, not as analogies to the construction [59].

3 The Chio Substrate

Chio is a capability-scoped mediation and evidence system. Signed artifacts use canonical JSON and fail closed on malformed or mismatched hybrid objects. The specification names what is outside scope: no global consensus, no public certification market, no arbitrary plugin trust expansion, and no theorem-prover coverage beyond the named assumptions in §9.

Capabilities and attenuation. A capability token names an issuer, subject, scope, expiry, and optional attenuation proof. Scope contains grants over tool servers, tool names, operations, and constraints. Delegation is valid only when the child scope refines the parent; budget splitting is separately constrained by sibling-sum soundness. The protocol maps those claims to proof identifiers such as `proof.delegation_step_allow_requires_attenuation` and `theorem.budget.sibling_sum_soundness`. No polity member can widen a delegated grant by relabeling it as constitutional authority.

Receipts. Every mediated call yields a signed receipt body. The body includes the capability id, tool server, tool name, canonical action hash, decision, policy hash, guard evidence, optional metadata, tenant id, trust level, and kernel public key. The protocol binds signed payloads to RFC 8785-style canonical JSON [50]; Chio’s local proof tree contains receipt integrity entries including `proof.receipt_sign_then_verify` and `proof.receipt_immutability`. Receipts become the substrate’s state transition facts.

A receipt is a signed statement about an attempted act at a trust boundary, with semantics closer to a court record than to a log line: it records which capability was presented, which tool action was evaluated, which policy hash was in force, which evidence was collected, which decision was made, and which kernel key signed the result. The sovereignty construction uses this property: a polity is a predicate over which signed receipt statements it will admit into its own history.

This framing removes a class of category errors. If a vendor signs a receipt, the receipt is evidence that the vendor’s kernel made a decision. It is not evidence that a buyer accepts the decision, that a second vendor accepts the lineage, or that a treaty admits the action. Admission is an additional check performed by the receiving polity (Figure 1 sketches the pre-dispatch admission flow). The receiving side may require the sender’s receipt, a continuation token, a bilateral invocation statement, a ladder-intersection artifact, selective-disclosure proof, and a public-witness bundle. Each artifact can be valid in isolation and still fail admission if it does not bind to the same receipt graph.

Bilateral DSSE. Cross-organization actions use a strict Chiodos predicate profile over a DSSE envelope [52]. The protocol pins `PREDICATE_TYPE_CHIODOS_BILATERAL` to `chio.bilateral-cosign-invocation.v1` and defines a treaty binding reference with treaty id, treaty-scope hash, ladder-intersection hash, admission report hash, continuation hash, request and outcome hashes, local and remote receipt hashes, leases, governance references, and signer kernel ids. The verifier requires exactly two independent signer keys for bilateral mode and rejects noncanonical payloads, wrong predicate types, stale leases, missing governance receipts, and subject-digest mismatch.

The protocol is stricter than generic in-toto-style provenance. In-toto records supply-chain steps and materials [61]; SLSA sets provenance levels for build integrity [58]. Chiodos uses those ideas, but the missing primitive for agent federations is joint authorization intent. A bilateral invocation says: two kernels saw the same treaty-bound action, over the same request and outcome digests, under the same continuation and receipt hashes, and both signed that predicate. Without that predicate, a receiver can verify that some party signed something, but cannot distinguish a shared action from two adjacent local logs that happen to name the same workflow.

The subject digest is over the canonical receipt body, not a presentation-specific byte stream. That design means a verifier can reject whitespace, field-order, or envelope rewrites without changing the semantic receipt identity. It also means privacy projections and public-witness records can refer to the same underlying subject without revealing every field. The cost is rigidity: schema evolution must preserve explicit adapters, and noncanonical legacy receipts need migration or rejection rather than informal acceptance.

Trust ladder. The Chiodos ladder has five ordered modes: observation, guarded, receipt-backed, partition-contingency, and quorum-required. The Rust rank function (`ladder_mode_rank`) maps those modes to increasing integers in `crates/chio-chiodos-runtime/src/treaty.rs:681`. Treaty intersection picks the higher-intensity requirement per action class and rejects destructive actions below the receipt-backed floor. Unknown action classes default to denial rather than routine admission.

The ladder is not a user-interface severity label. It is a join rule for multi-polity admission. If one participant treats an action as ordinary and another participant treats it as quorum-required, the treaty intersection must adopt the stricter requirement or deny. This is why the ladder belongs inside the treaty artifact rather than only in local policy files. The receiver needs to know not just that the sender's local policy allowed the action, but that the joint action class was evaluated at the jointly required mode.

The five modes encode strictly increasing evidence demands. Observation admits read-only telemetry whose action consequences are weak. Guarded requires local policy mediation but no cross-kernel evidence. Receipt-backed requires signed receipt evidence for the action class, and destructive actions are rejected outright below this floor. Partition-contingency reserves explicit reconciliation rules for cases where partitions are expected and the receiver must record a reconciled history. Quorum-required demands multi-party co-signing, with the joint co-sign rank computed against the participant manifests. Ladder stability is proved over this finite ladder; a sixth mode would require coordinated changes to the schema, Rust validators, and proof model rather than a convention on integer ordering (§9).

Selective disclosure. The selective-disclosure crate projects receipt, workflow, and step records into BBS message vectors. The BBS feature pins a SHA-256 ciphersuite string, signs projections, derives disclosure proofs, and verifies proofs against a verifier-owned issuer registry. The spec keeps Ed25519 over canonical receipt bytes authoritative; BBS is a secondary commitment for disclosure, not a replacement signature [7, 9, 38].

Selective disclosure makes treaty evidence partial by design. A buyer may prove that a vendor action was receipt-backed, within

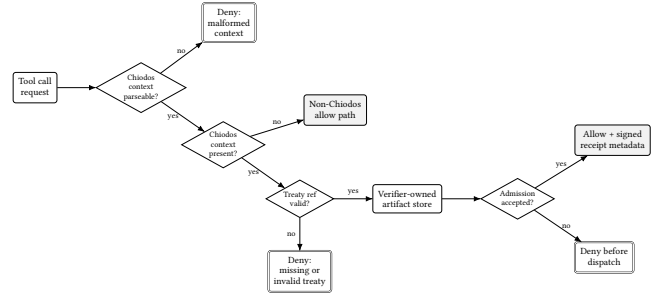


Figure 1: Pre-dispatch admission. A federated request without valid treaty context is denied before tool invocation.

budget, and under a live treaty scope while withholding tool arguments, tenant identifiers, or unrelated evidence fields; a regulator may require a different projection of the same receipt. Chio's projection model declares which receipt fields are wholesale-only and which can be disclosed selectively, with the verifier pinning issuer keys and projection versions so the proof is not free-form redaction. Bilateral treaty admission across organizational boundaries requires evidence without total surveillance, and the BBS proof carries exactly the predicate the receiving polity needs.

Anchoring. The anchor crate normalizes local inclusion proofs into multi-lane evidence. Its module comment lists EVM root publication, checkpoint-to-Bitcoin super-root aggregation with OpenTimestamps, Solana memo records, and fail-closed proof bundle verification. The Rekorder lane verifies signed-entry timestamps against pinned Rekorder keys but still labels Merkle inclusion-proof verification as follow-up. The OTS Bitcoin lane parses timestamp evidence but fails closed for load-bearing public-witness policy until trusted Bitcoin header or calendar-backed commitment evidence is in the receipt contract. That limitation is retained in Section 9.

Tile-based transparency-log designs (Trillian Tessera and the transparency.dev witness network) are the substrate the anchor crate's Rekorder lane is converging on as inclusion-proof verification moves out of follow-up status [62]. Adopting tile-log architecture imports a tile-boundary inclusion-proof attack surface (a witness signing tile N consistent with two distinct tile- $(N+1)$ extensions that share the boundary leaf set), which the present construction does not address; a per-anchor-epoch tile-boundary-consistency proof obligation is left as future work. Anchoring is treated as a witness service, not as an oracle of legal truth. A public witness can make equivocation more expensive by pinning a root outside the polity's local store. It cannot prove that the original policy was wise, that a human contract was valid, or that a third-party court will enforce anything. This is the same discipline applied to the rest of the system: external evidence strengthens auditability only when the verifier states exactly which statement it proves and exactly which residual assumptions remain.

4 Programmable Sovereignty: Formal Model

The model abstracts away networking, cryptography, and storage; those lower layers are audited assumptions in the proof manifest.

What remains is the institutional core, expressed as decidable predicates.

Trace semantics. Let R be the set of receipt identifiers and E the set of evidence artifacts addressable by digest. A local execution trace is a sequence

$$\sigma = \langle (r_0, E_0, d_0), \dots, (r_n, E_n, d_n) \rangle$$

where each $r_i \in R$ is a receipt, $E_i \subseteq E$ is the evidence set presented at admission, and $d_i \in \{\text{allow}, \text{deny}\}$ is the receiving kernel's decision. A polity P accepts σ iff every allowed element satisfies P 's admission predicate and every denied element carries a failure code permitted by the constitution. Denial is part of the constitutional record: a polity that suppresses denials can no longer account for which attempted acts were rejected at its boundary.

Access control asks whether a call may proceed; constitutional admission asks whether the polity accounts for it. The separation is what makes cross-kernel events first-class: a rejected continuation, a stale treaty, a missing co-signature, or a privacy-proof mismatch is a constitutional event even when no tool ran.

Polities. A polity is a triple (T, C, K) where T is a *PolityScope* (a closed predicate over receipt identifiers), C is a Merkle-rooted citizenship roster, and K is a constitution given as a finite list of predicates, each backed in production by a Rust check symbol and a theorem-inventory entry. In the Lean paper model:

$$\text{polityAdmits}(P, r) = P.\text{scope}(r) \wedge \text{allPredicates}(P.\text{constitution}, r).$$

A polity's authority is therefore strictly local: it can accept or reject receipts at its own log boundary. Territory is the signed hash-pinned tuple over capability namespaces, receipt schemas, ladder manifests, and denied predicates, not land.

The citizenship roster C is similarly weak. It gives a finite set of keys, kernels, or authorities standing inside a receipt-admission scope; it does not grant personhood or political rights in the human sense. A constitution may bind amendment proposals to citizen quorum, require citizen attestation for treasury exits, or admit a foreign kernel into treaty standing without citizenship. Each rule is a predicate over receipt admission and governance artifacts.

The polity's authority extends to refusing foreign evidence until the treaty predicate is satisfied; that refusal is the programmable surface. It does not extend to forcing another polity to accept a receipt, compelling a public witness to publish a root, or obliging a state court to recognize a judgment.

Admission. The admission relation $(s, r) \models (T, K)$ holds iff r is in scope T and every constitutional predicate in K accepts on (s, r) . The runtime counterpart is pre-dispatch: `crates/chio-chiodos-runtime/src/admission_hook.rs:88` (`fnevaluate`) routes admission, denying malformed Chiodos admission context on the early-return branch and federated origin without treaty context at :168; on a parseable context it calls the runtime admission evaluator before returning allow or deny. The relation is decidable: predicates are finite, the schema uses concrete discriminants, and validation returns a Boolean or an error mapped to deny.

Fail-closed behavior is a semantic requirement, not an engineering preference. If parsing an admission artifact can fail open, the parser becomes an unrecorded legislature that admits receipts outside the predicate set. The model therefore treats every validation

error as a denial with evidence: the denial receipt must be available to auditors so that failure modes do not disappear from the polity's history.

Attenuation. Capability delegation is a refinement relation. If cap' descends from cap , then every invocation allowed under cap' is allowed under cap with equal or stricter budget and lifetime. The proof inventory names this surface with `proof.delegation_step_allow_requires_attenuation`, `proof.delegation_step_allow_requires_expiry_monotonicity`, and `theorem.budget.sibling_sum_soundness`. Sibling-sum soundness is the economic counterpart: two child delegations cannot jointly spend more than the parent share.

Attenuation gives the model an audit property. A polity member can delegate narrower authority without losing the ability to audit the child path against the parent grant. If the child path exceeds the parent's scope, the proof obligation fails at the level of admitted receipt semantics; if the child path stays within scope but a budget allocator permits sibling overspend, treasury predicates can be bypassed. Budget soundness therefore belongs in the constitutional model, not in an accounting appendix.

Treaties. A bilateral treaty τ between polities P_1 and P_2 is a pair (T_τ, K_τ) together with references to both parties' constitutions; Figure 2 sketches the bilateral handshake. The Lean theorem `treaty_admission_iff_predicate_intersection` states:

$$\forall \tau, r. \text{treatyAdmits}(\tau, r) \Leftrightarrow \text{treatyPredInter}(\tau, r).$$

Here `treatyPredInter` is the Boolean conjunction of treaty scope, treaty constitution, and both polities' *polityAdmits* relations restricted to their scopes. The proof in `formal/lean4/Chio/Chio/Treaty/Intersection.lean:111` cases on the treaty record, then on each polity, unfolds the four definitions *treatyAdmits*, *treatyPredicateIntersection*, *polityAdmits*, and *constitutionAllows*, and discharges the resulting Boolean equality by associativity. The corollary is that admission cannot be widened by re-arranging conjuncts: any sound implementation that returns true on `treatyAdmits` must also return true on `treatyPredInter`. This is the load-bearing theorem for cross-kernel admission, since it pins both polities' independent decisions to the same Boolean conjunct set. The production counterpart at `crates/chio-chiodos-runtime/src/treaty.rs:264` validates treaty scope, freshness, manifest coverage, declared participant hashes, action-class coverage, mode joins, destructive floors, and co-sign ranks before any required evidence is accepted.

Treaty admission is not federation by name. Two polities can list each other as participants and still deny every cross-boundary action if their predicates fail to intersect. A treaty can admit a narrow action class without merging citizenship rosters, treasuries, policy files, or receipt stores. The shape is closer to a typed interface than to a social graph.

Ladder stability. The theorem `treaty_admission_stable_under_ladder_floor` states:

$$\forall \tau, m, r. m.\text{atLeast}(\tau.\text{modeFloor}) \Rightarrow \text{treatyAdmitsUnderMode}(\tau, m, r) = \text{treatyAdmits}(\tau, r).$$

The proof unfolds *treatyAdmitsUnderMode*, rewrites the leading conjunct using the floor hypothesis, and applies Boolean reduction. The consequence is that ladder checks compose with the treaty predicate without changing its decision: a presented mode at or

above the floor is invisible to admission, and a mode below the floor short-circuits to deny without even consulting the predicate. This is also load-bearing for cross-kernel admission, because it guarantees the ladder-floor and the treaty predicate cannot disagree on admissibility once the floor is satisfied. The production counterpart appears in the mode-rank check at `treaty.rs:681` (`ladder_mode_rank`) and the co-sign-rank check at `treaty.rs:706` (`co_sign_requirement_rank`), with the floor enforcement consolidated in `validate_ladder_intersection` at `treaty.rs:407` (all under `crates/chio-chiodos-runtime/src/`). The theorem is bounded to the current five-mode ladder; a sixth mode forces coordinated changes to schema, validators, and proof model (§9).

Threat model. The adversary can control the agent process, choose malformed tool arguments, replay old treaty references, omit evidence, present receipts from a different treaty, reorder JSON fields, use stale ladder manifests, reuse signer keys, or attempt to smuggle trust roots through request metadata. The adversary may also operate an honest-looking but non-admitted foreign polity, or attempt to inject pins, revocation entries, or peer keys into a verifier’s trust store; provisioning of those stores is itself a treaty-admitted, governance-receipted action and is outside the adversary’s reach by assumption. The adversary cannot break Ed25519, BBS, SHA-256 collision resistance, or the correctness of canonical JSON within the assumptions in Table 4. The adversary cannot force the verifier to use sender-owned trust stores because the runtime checks verifier-owned store artifacts. Two kernels operated by a single actor satisfy two-key DSSE but not party-independence, which the present construction treats as an operational-discipline assumption (§9).

The model discharges three obligations: single-amendment backward refinement (as a type-level invariant on *enactAmendment*), bilateral treaty admission as predicate intersection, and ladder-floor stability of treaty admission. Three downstream obligations follow that the model does not discharge and that the substrate makes load-bearing in a federated deployment. Trajectory invariants for amendment: a constitutional-ratchet attack composes individually valid refinements into an adversary-only predicate, and ruling it out requires an essential-predicate invariant in addition to per-step refinement. Implementation refinement: a constant-time admission ladder, finite-arithmetic budget proofs, and wire-uniform error responses are protocol-level requirements whose discharge belongs to the Rust-Lean refinement program rather than to the bounded model (Section 9). Treaty composition across more than two participants: associativity and acyclicity of composed treaties become load-bearing when the graph admits cycles, and an external anchor witness is the substrate primitive that closes them.

The model does not solve byzantine consensus across all polities. It solves the problem of a receiver deciding whether to admit a receipt under its own constitution and treaty predicates. If two polities fork their histories, Chio exposes the fork and makes future admission conditional on reconciliation evidence; it does not impose a single world history without an additional consensus layer.

Amendments. An amendment from K to K' is admissible only with a backward-refinement witness:

$$\forall r. \text{allPredicates}(K', r) \Rightarrow \text{allPredicates}(K, r).$$

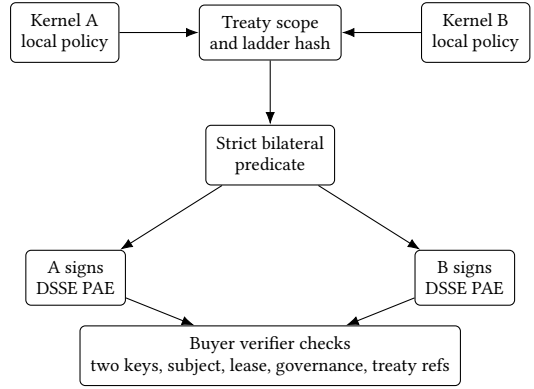


Figure 2: Bilateral treaty admission. Both kernels sign the same strict predicate over receipt and treaty bindings.

This is the “votes on theorems, not bytes” rule (Figure 3). A vote may approve a proposed constitution, but enactment is rejected when the delta proof fails. The invariant is type-level rather than runtime-level: *enactAmendment* takes a *ConstitutionalDelta*, which by construction carries *proofTerm : BackwardRefines new old*. A verdict claiming enactment without the refinement witness cannot be constructed by the well-typed runtime path, because *enactAmendment* does not type-check absent the delta. Two named theorems make the structure explicit: *amendment_admissible_iff_backward_refinement* records the iff between *amendmentAdmissible* and *BackwardRefines* (definitionally, by *rfl*), and *amendment_without_refinement_rejected* establishes that the independent *rejectAmendment* path produces *AmendmentVerdict.rejected*. The substantive content shifts from a Boolean check on a trusted caller to a type-level requirement that enactment carry the refinement proof (formal/lean4/Chio/Chio/Treaty/Intersection.lean:84 for the *ConstitutionalDelta* structure and :103 for *enactAmendment*). The two theorems are definitional refinement bridges rather than load-bearing proofs; their work is to anchor the type-level invariant in the proof inventory.

Backward refinement is conservative by design. Many real constitutions permit discontinuity: revolutions, emergency powers, amnesties, retroactive statutes, judicial reinterpretation, debt restructuring, expulsion, founder override, regulator preemption, treaty repudiation. Chio represents such events only as explicit crisis artifacts. The normal amendment path requires K' to imply K on receipts already admitted under K , and auditors verify mechanically whether a new constitution narrows admission, preserves already-admitted history, or introduces an unproved break.

5 Implementation

The implementation has six load-bearing components: a runtime kernel that exposes a pre-dispatch admission hook and a treaty primitive; a federation layer that performs strict bilateral DSSE verification; a selective-disclosure module over a BBS message vector; a multi-lane anchor for public-witness evidence; a fixture-backed three-vendor case-study generator; and the Lean proof root. Each implementation citation below resolves to a `file:line`, and each citation falls into one of three categories: a runtime obligation that

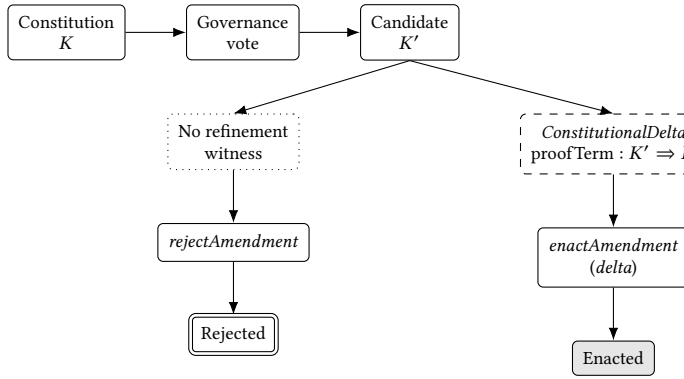


Figure 3: Amendment lifecycle. Enactment is type-conditioned on a *ConstitutionalDelta* carrying a refinement witness ($\text{proofTerm} : K' \Rightarrow K$); without one, the only constructable verdict path is *rejectAmendment*. The split is structural, not a runtime decision: the no-delta enact path does not type-check.

discharges a Lean theorem, a tested executable path, or a fixture. Table 1 indexes the artifacts; Table 2 maps them to Lean theorems.

Treaty primitive. The treaty module `crates/chio-chiodos-runtime/src/treaty.rs` discharges the runtime side of `treaty_admission_iff_predicate_intersection`. It hashes treaty scope with canonical JSON, computes semantic hashes that exclude validity windows for stable intersection identity, and binds bilateral invocations over schema, invocation id, treaty id, ladder-intersection hash, continuation hash, action class, consistency model, capability id, request hash, outcome hash, local and remote receipt hashes, and signer ids (`crates/chio-chiodos-runtime/src/treaty.rs:84`). `validate_treaty_scope` rejects unsupported schemas, invalid windows, missing participants, duplicate participants, hash mismatches, and malformed trust bundle references (`crates/chio-chiodos-runtime/src/treaty.rs:191`). `compute_ladder_intersection` then checks participant manifests and constructs the joint action-class table.

The treaty module turns political metaphors into ordinary data validation. A treaty with one participant rejects; with duplicate participant keys, rejects; with a governance ladder hash unequal to the canonical hash of the manifest, rejects; with an action class absent from one participant’s manifest, rejects; with a destructive action whose joined mode falls below receipt-backed, rejects. A programmable constitution fails as a schema-and-predicate matter before it becomes a governance debate.

The semantic-hash split separates stable identity from liveness. Treaty-scope semantic identity excludes the validity window so that short-lived renewals carry the same policy identity while admission still enforces freshness. Stable identity belongs to the predicate set; liveness belongs to the admission context.

Admission hook. `crates/chio-chiodos-runtime/src/admission_hook.rs` implements `RuntimeAdmissionHook`. It permits non-Chiodos requests only when no Chiodos runtime context is

present; otherwise malformed context denies. Federated origin without treaty context denies with `missing_chiodos_treaty_context`. Request-level smuggling of trust roots, peer directories, signing keys, or dynamic trust bundles is rejected by `treaty_ref_from_request` (`crates/chio-chiodos-runtime/src/admission_hook/treaty_ref.rs:23`). Treaty DSSE evidence is verified against verifier-owned store artifacts; signer sets must equal treaty participants and public keys must be independent (`crates/chio-chiodos-runtime/src/admission_hook/dsse.rs:76`). The hook releases reserved continuation state after runtime denial or abort, closing a replay edge tested in `crates/chio-chiodos-runtime/tests/runtime_admission.rs`.

The hook is the point at which the model is falsifiable against real code. Allowing federated origin without treaty context would make the treaty-intersection theorem irrelevant to cross-kernel execution. Allowing request metadata to smuggle a trust root would collapse the verifier-owned-store assumption. Failing to release reserved continuation state on denial would create a state leak and ambiguous replay behavior. Each of these is a concrete place where agent middleware turns formal policy into advisory logging.

The hook is asymmetric: non-Chiodos local requests use the old path when no Chiodos context is present, and once Chiodos context appears, malformed or incomplete context denies. Backward compatibility is preserved; partial federation is not a bypass. The receiver’s kernel decides when a request has crossed into the constitutional path.

Strict DSSE verifier. `crates/chio-federation/src/bilateral_dsse.rs` defines the strict predicate and envelope used by every cross-organization invocation. `verify_chiodos_dsse_envelope` rejects wrong payload type, wrong signature count, noncanonical JSON, wrong statement type, wrong predicate type, missing subject, and reused signer keys (`crates/chio-federation/src/bilateral_dsse.rs:996`). `crates/chio-federation/src/bilateral_verifier.rs` layers operational checks on top: peer pins, fresh ladder manifests, revocation epoch, receipt resolution and signature, tool argument hash, policy verdict agreement, capability lease, governance receipt for receipt-backed classes, and treaty binding refs (`crates/chio-federation/src/bilateral_verifier.rs:859`). Together, the envelope check and the operational check ensure that admission depends on the predicate the protocol names, not on a strict subset that a sender could satisfy with degenerate data.

The strict verifier refuses “signed JSON” as an adequate security boundary. A DSSE envelope with the wrong predicate is not a near miss; a payload signed by the same key twice is not two-party consent; a receipt with a valid signature but the wrong request hash is not admissible under the treaty; receipt-backed classes require a governance receipt even when the ordinary receipt is otherwise valid. The error taxonomy is part of the protocol because downstream audit must know which constitutional precondition failed.

Buyer audit and three-vendor loop. The three-vendor example is a generator in `examples/chiodos-3vendor/src/main.rs` that emits buyer-auditor proof packages, workflow-intersection JSON, signed negative packages, pheromone artifacts, and runtime-spine fixtures. The example is gated by integration tests that re-run the closure on every change to any of its bound components.

Table 1: Load-bearing implementation artifacts used by the paper.

Artifact	Enforced statement	Evidence boundary
TreatyScope	Participants, public keys, ladder-manifest hashes, allowed action classes, validity window, revocation epoch, and trust bundle are bound before intersection.	Schema and semantic hash checks are in the runtime; treaty law outside these fields is not represented.
LadderIntersection	For each allowed action class, the joint mode, consistency model, co-sign rule, evidence set, and per-participant mode are materialized.	Correct only for the finite ladder schema accepted by the current validators.
CrossKernelContinuation	Cross-kernel child work is linked to parent receipt, session anchor, capability id, action class, target tool, nonce, and validity window.	Prevents replay only when the receiving runtime consumes or releases continuation state correctly.
BilateralInvocation	Local and remote receipts, request and outcome hashes, lineage statement, continuation, action class, consistency model, and signer ids are jointly bound.	Verifies joint intent; it does not prove the underlying human commercial contract.
SelectiveDisclosureProof	A verifier can check disclosed projection fields against an issuer key without receiving every receipt field.	Privacy proof is secondary to the authoritative Ed25519 receipt.
AnchorWitnessPolicy	Public witness lanes must verify active inclusion or use explicitly cached verifier-owned evidence.	Rekor SET verification is stronger than OTS advisory mode; lane-specific limitations remain.

Buyer closure is the concrete example of scoped sovereignty. The buyer demands the specific evidence package required for admission, not every internal receipt; the package binds local and remote receipts to treaty scope, ladder intersection, continuation, and lineage. A system that cannot generate signed denials cannot prove it rejects stale, mismatched, or replayed evidence; the generator produces both positive and negative packages in the same canonical schema.

The example demonstrates that the runtime and federation crates agree on artifact shape, emit repeatable fixtures, and exercise negative cases. Because every bound component (treaty scope, ladder intersection, bilateral DSSE, admission hook, buyer audit) is exercised, a regression in any one breaks the integration gate.

Proof inventory. The Lean proof root contains the theorem and lemma declarations the runtime depends on, including the ones introduced for this work, with no admitted goals (sorry) and no extra axioms (axiom). A curated inventory bridges formal names to protocol properties and Rust symbols; the curated subset is the audited coverage the paper defends.

Cryptographic primitives, canonical-JSON implementation correctness, TLS, SQLite, public-chain finality, and several other lower layers are outside the Lean proof root (Table 4). Selected constitutional predicates and protocol invariants have named Lean theorems, no unchecked placeholders, and inventory entries linking those theorems to protocol properties.

6 Evaluation

Each reported number is produced by a script in bench/; absent measurements are marked rather than estimated. The baseline machine is a MacBook Pro 18,4 (Apple M1 Max, 10 cores, 32 GB RAM) running Darwin 25.4.0.

Scripts live at papers/programmable-sovereignty/bench/ and write TeX results under bench/results/ alongside the raw log or CSV. Hand-written harnesses compile temporary Rust projects against the source tree referenced by CHIO_SOURCE (set to a workspace checkout, or absent for the repository root); the dispatch and replay scripts redirect CARGO_TARGET_DIR into a temporary directory so measurement does not mutate the source tree. The four exercised paths are dispatch admission, treaty intersection, selective-disclosure verification, and replay-corpus stability (Table 3).

Treaty intersection. The treaty-intersection harness constructs two governance ladder manifests with N overlapping action classes,

calls `compute_ladder_intersection`, and reports per-iteration percentiles. The exercised paths are real: schema checks, freshness checks, manifest hash binding, destructive floor handling, mode ranks, consistency model validation, and co-sign rank joins. Growth is approximately linear in N because the implementation walks every allowed class and every participant manifest, hashes manifests, and builds per-class joined state. The p99 outlier at $N=100$ is reported, not smoothed.

Selective disclosure. The selective-disclosure harness enables the bbs feature, constructs a real `ChioReceiptBody`, projects it into a BBS vector, signs the projection, derives a disclosure proof for three fields, verifies it against an issuer registry, and records proof size and verification latency. This measures the opt-in privacy path, not the authoritative Ed25519 receipt signature.

BBS verification is slower than dispatch and treaty intersection by orders of magnitude; the cost is pairing-heavy cryptography in a debug-profile harness. The selective-disclosure path is executable end-to-end, has a measurable proof size, and is suitable for audit or treaty-boundary insertion rather than per-dispatch use.

Dispatch. The dispatch script reuses the production Criterion benches for `dispatch_allow`. The receipt-sign and receipt-append benches are excluded: their bodies black-box a constant, so reporting them would measure Criterion overhead, not Chio.

Receipt-shaped benchmark scaffolds exist whose bodies black-box a constant; reporting their output would manufacture a number. Receipt signing, receipt verification, and anchor inclusion require dedicated measurement harnesses before any latency claim becomes admissible (§9).

Replay. The replay script runs the kernel-capability golden target with `CHIO_BLESS=0`. Locally this validates 50 fixtures distributed with the runtime, across ten capability-side families (allow simple / metered / with-delegation; deny expired / revoked / scope-mismatch; guard-rewrite; replay-attack; tampered canonical-JSON; tampered signature) via `cargotest-pchio-replay-gate--testgolden_byte_equivalence`. The reported test wall-time is 1.07 s; surrounding workspace compilation dominates wall-clock when run cold. The test asserts *manifest-stable across machines*: each fixture’s expected verdict is canonicalised against a stored golden manifest, so the property is reproducibility of canonical verdicts rather than re-execution of the kernel on adversarial input. Additional fixture families (corpus smoke, replay proptests, cross-version replay) are exercised separately.

Table 2: Selected proof-to-code map. The first four rows are theorems introduced in this work.

Theorem	Informal statement	Code or property
treaty_admission_iff_predicate_intersection	Treaty admission equals treaty, left-polity, and right-polity predicate intersection.	crates/chio-chiodos-runtime/src/treaty.rs:264
treaty_admission_stable_under_ladder_floor	Satisfied ladder floor preserves treaty admission.	crates/chio-chiodos-runtime/src/treaty.rs:407 (validate_ladder_intersection)
amendment_admissible_iff_backward_refinement	Amendment admission iff new predicates imply old predicates.	Polity amendment model
amendment_without_refinement_rejected	Missing proof rejects enactment.	Crisis admission invariant
proof.delegation_step_allow_requires_attenuation	Delegation allow implies attenuation.	Capability delegation
theorem.budget_sibling_sum_soundness	Child budget sum stays within parent share.	BudgetSplit::try_admit_child
proof.receipt_sign_then_verify	Signed symbolic receipt verifies.	Receipt signing
proof.verified_receipt_lineage_sound	Verified lineage requires both receipts and linkage.	Buyer lineage audit

Table 3: Measurement status on the baseline machine. Withheld rows have a target surface but no dedicated measurement harness; the cells are marked rather than estimated.

Metric	Script	Result	Status
Kernel dispatch allow	bench/run-dispatch-allow.sh	p50 72.051 us (Criterion CI 71.745 us to 72.367 us)	measured
Treaty intersection p50/p99	bench/run-treaty-intersection.sh	N=1 p50 131.67 us p99 194.00 us; N=10 p50 539.75 us p99 792.00 us; N=100 p50 4980.46 us p99 12138.25 us	measured
Selective-disclosure proof size and verify p50/p99 (BBS)	bench/run-selective-disclosure.sh	624 bytes, p50 161824.17 us, p99 679380.12 us	measured
Receipt sign p50/p99	(none)	[unreported]	withheld
Receipt verify p50/p99	(none)	[unreported]	withheld
Anchor inclusion per lane	(none)	[unreported]	withheld
Replay corpus manifest stability	bench/run-replay-corpus.sh	50 fixtures, manifest-stable across machines (test wall-time 1.07 s)	measured

The 50-fixture corpus does *not* exercise the Chiodos buyer-closure paths the rest of this paper defends: zero fixtures touch the strict bilateral-DSSE rejection codes named in §5, the constitutional-ratchet attack class, or the multi-lane anchor witness-quorum surfaces. Those families are explicit follow-up work for a Chiodos-specific replay corpus (§9).

Case study. In the three-vendor loop, the buyer owns the verifier context and proof package. Vendor receipts are not imported merely because they are signed; they must bind treaty scope, ladder intersection, continuation, lineage, bilateral invocation, and strict DSSE evidence. The admission hook consumes a treaty continuation before dispatch and releases it on denial; a replayed continuation is denied, which makes buyer audit an enforcement path, not a report-generation path.

This construction inverts ordinary audit export. Ordinary export hands the buyer a bundle after the fact and asks for trust; the Chiodos path puts the buyer’s verifier context into the admission decision, so stale leases, wrong policy verdicts, missing governance receipts, and mismatched request hashes deny before the receiving action runs. The primary security property is pre-dispatch admission under buyer-owned predicates; the after-the-fact audit package becomes corroborating evidence.

The buyer-closure generator emits signed positive packages and signed negative packages. A positive package witnesses an admitted vendor-to-vendor action with treaty scope, ladder intersection, continuation, lineage, bilateral invocation, and strict DSSE evidence bound to one canonical subject digest. A negative package witnesses a denied path (stale lease, mismatched request hash, replayed continuation, missing governance receipt) in the same canonical schema. Every admission decision the buyer made is replayable; every denial is auditable in the same format as an allow; and a downstream reviewer replays the loop without privileged access to vendor internals.

Reproducibility hazards. Measurements are machine-local; generalization requires fresh runs on stated hardware. The dispatch number uses a Criterion sample count chosen for reproducibility on the baseline machine. The hand-written harnesses use debug builds and serve as transparent executable checks, not tuned microbenchmarks; release-profile re-runs are noted as a limitation in §9.

7 Discussion

Chio instantiates receipt-bounded polities whose laws are executable predicates and whose external relations are admission contracts. A treaty scope type, ladder-intersection artifact, cross-kernel continuation, strict bilateral-invocation binding, fail-closed runtime admission hook, buyer attestation review package, BBS selective-disclosure mechanism, multi-lane anchor, and bounded Lean model for treaty intersection and amendment refinement realise that property.

The polity claim is declarative, not Westphalian. A Chio polity has final authority over its receipt-admission boundary; it has none over physical persons, land, or external courts. Its territory is a receipt namespace, its citizens are DIDs or kernels admitted into that namespace, its border is the runtime admission hook, its courts are verdict artifacts over receipts, and its diplomacy is recognition of peer predicates rather than diplomatic immunity. Montevideo Article 1 lists permanent population, defined territory, government, and capacity to enter relations; Article 3 says political existence is independent of recognition by other states [15, 53]. The formal model exposes analogs for each criterion (citizenship roster, receipt-scope territory, constitutional predicates, treaty intersection) but does not claim the human territorial reading. The constitutive-recognition pushback against Article 3 also stands: most diplomatic practice uses constitutive recognition, and Chio cannot supply it. The regulator-facing sentence is therefore that a Chio polity is a

machine-checkable governance committee over a declared receipt log, optionally wrapped by a legal entity.

The deeper framing is Hartian, and the scope of the claim matters. Hart distinguished three conditions for a rule of recognition to exist as law: it must be a criterion officials apply to identify which rules count as law of the system (*The Concept of Law*, p. 94), it must be applied as a settled practice (p. 110), and officials must accept it from the internal point of view as a common standard (p. 116). Chio’s admission predicate K instantiates the first condition concretely: a machine-checkable test that any party can replay over the same canonical bytes to determine whether a receipt belongs to the polity’s history. The second and third conditions are sociological obligations Chio does not discharge – kernels execute K deterministically, which is mechanical rather than practice, and operators accept K as a software dependency rather than as authority. A candidate analog for Hart’s officials is the cosigner network and its key-publication / denial-honoring behavior, but the present construction does not develop that analog. Raz’s authority-as-content-independent-reasons-for-action [48] maps onto capability tokens whose validity is constitutive of the authority rather than reasons for obeying it, and Schauer’s rules-vs-standards distinction [51] maps onto predicates-vs-judicial-discretion with Chio at the extreme rule-bound end – the Dworkin critique on principles, where no predicate decides and judicial reinterpretation is needed, applies in full. The construction is therefore not a new theory of sovereignty but a positivist contribution narrower than the term suggests: a constructive instance of *condition (a)* of the Hartian rule of recognition [30] over cross-organizational tool invocations.

External jurisdictions remain asymmetric. The design models the EU as a Supranational external peer by stacking GDPR or procurement predicates on admission; the asymmetry is that those predicates bind Chio’s kernel, not the EU’s. The relation becomes bilateral only when an external institution supplies a DID, a trust bundle, and a co-signing key. Until then, integration is one-way compliance evidence, which is the input the substrate is engineered to produce. The relevant regulatory text is now finalized for portions of this surface: the EU AI Act General-Purpose AI Code of Practice came into force in August 2025 with transparency and safety chapters that name the logging and incident-record obligations Chio’s receipts are shaped to discharge [25], and the NIST AI RMF Agentic Profile defines a parallel set of obligations for the US federal supplier surface [12].

An enterprise can encode procurement, privacy, or sanctions predicates without asking a regulator to run a Chio kernel, and can produce signed evidence that it refused nonconforming receipts. The evidence is auditable in the same canonical schema as the admitted receipts it accompanies, and the relation becomes bilateral when the external institution participates cryptographically with its own DID and co-signing key.

Próspera, FTX, and Tornado Cash mark the failure modes the construction is designed against. Próspera shows that charter-city autonomy can collapse into sovereign conflict and arbitration. FTX shows that exchange governance theater without verifiable custody and receipts is not institutional discipline. Tornado Cash shows that immutable code and governance artifacts can exceed statutory categories. Chio’s answer is not immunity but locality: every asserted authority is local, typed, signed, and attackable.

On-chain governance decides bytes, parameters, or social mandate; Chio’s amendment path decides whether the new predicate set refines the old one for admitted history. A vote may approve a candidate constitution; enactment is gated on a Lean term, and a vote that overrides the gate is recorded as a crisis artifact rather than absorbed silently. This is the load-bearing difference between programmable governance as commonly deployed and proof-carrying constitutional change.

The economics layer is downstream of admission. Chio ships market, settlement, credit, underwriting, and anchoring components, but programmable sovereignty does not depend on an endogenous token or market before admission semantics are sound; the ordering is authority first, receipts second, settlement third. A future interoperable profile needs the receipt fields, canonicalization rules, treaty-binding digest, and rejection codes any receiving organization must require before admitting a foreign tool call.

The construction is complementary to pre-deployment AI safety work. Capability evaluations (METR, UK AISI Inspect, ARC Evals) and frontier-safety frameworks (Anthropic RSP, OpenAI Preparedness, DeepMind Frontier Safety) bound what a model is likely to do; they produce no tamper-evident record of what a deployed agent actually did across a vendor boundary. Alignment-faking [28] and situational-awareness [6] research sharpen this gap: a model that behaves under evaluation and differently in deployment leaves no checkable artifact unless the runtime emits one. Chio’s receipt graph is that artifact. The treaty-binding digest pins joint action intent across two organizations, so a scheming agent cannot retroactively deny a cross-vendor call’s predicate set. Constitutional AI [4] conditions outputs at training; Chio gates whether they cross a trust boundary at deployment.

The adoption path is incremental. One enterprise runs receipt-backed admission for its own high-risk tools. It then adds selective disclosure so internal teams can prove policy compliance without exporting full tenant data. It then admits a narrow partner action under a bilateral treaty scope. Federated courts, treasuries, and public witnesses are downstream of those steps, not parallel to them, which keeps the hardest political claims downstream of executable evidence.

8 Related Work

Capability operating systems established the security lineage. KeyKOS and EROS treated capabilities as the primitive access-control substrate [29, 55]. Capsicum adapted capability discipline to UNIX processes [68]. seL4 showed how microkernel authority can be connected to machine-checked implementation proofs [34, 54]. Chio borrows the discipline but changes the object of control: the protected act is an agent tool call whose decision becomes a signed receipt.

The evidentiary target differs from classic operating-system capabilities. KeyKOS, EROS, Capsicum, and seL4 enforce authority inside a machine or kernel boundary; Chio emits a signed receipt at the same kernel call that enforces the capability, and a third-party verifier replays the same predicate the enforcer evaluated. The treaty binds local and remote receipts rather than only naming principals, which is the structural difference from per-machine capability discipline.

Object-capability languages and formalisms supply the vocabulary for attenuation. Miller’s thesis, Murray’s analysis, and Drossopoulou and Noble’s policy work all treat authority as an explicit, attenuable relation rather than a global permission table [22, 23, 40, 41]. EIP-7702 and ERC-4337 ship the capability-attenuation idea at scale: tens of millions of Ethereum smart accounts now carry session-keyed delegation with budget and expiry as on-chain policy [8]. Where these works treat attenuation as a program-internal property or as a chain-settled policy, Chio promotes each attenuation step to an artifact a non-participant can verify against a constitutional predicate, with no chain dependency for the admission decision itself.

Fuchsia’s capability routing work is another useful contrast [26]. It gives componentized software a disciplined way to route authority. Chio’s problem is less about component topology and more about cross-organization evidence: once a tool call crosses a vendor or buyer boundary, the verifier needs a receipt graph, not just local component routing.

The agentic-systems security literature is now formalizing the same authority-boundary failures. IsolateGPT proposes per-app isolation with information-flow gates for LLM-based agents [69]; SAGA distributes access-control tokens from a user-controlled Provider [60]; subsequent work names confused-deputy and authority-boundary failures as the load-bearing class [11]. The constructions compose with Chio rather than compete: an IsolateGPT-style gate is a local-policy predicate inside a Chio constitution; SAGA’s Provider distributes tokens while Chio places admission at the receiving kernel under a verifier-owned predicate. A TEE-attested kernel (Intel TDX, AMD SEV-SNP, NVIDIA H100 confidential compute) gives stronger online guarantees because the kernel itself is attested, while Chio’s receipt cosigning gives stronger offline auditability because admission decisions replay from canonical bytes; the two strengthenings are complementary rather than competing.

Verified systems work supplies the assurance pattern but not the governance object. CompCert proves semantic preservation for compilation [36]; VST proves C programs against separation-logic specs [2]; IronFleet combines TLA-style refinement with implementation proofs for distributed systems [31]; Project Everest applies F* to secure communication software [47]; Lean 4 supplies the theorem-prover and programming-language base used here [19]. Two production analogs are most informative. AWS Cedar verifies per-principal policy decisions inside a single authority, with a Lean specification, a differential-tested Rust runtime, and a verification-guided development loop [16, 20]. SampCert verifies differential-privacy sampling primitives with a 12,000-line Lean corpus deployed inside AWS Clean Rooms [18]. Cedar and SampCert establish a Lean-plus-Rust industrial pattern over individual primitives; Chio’s theorems range over admission across two kernels each holding its own constitution, with treaty intersection and amendment refinement as the additional obligations the single-primitive case does not face.

Transparency and supply-chain systems define adjacent evidence formats. Certificate Transparency, Rekor, DSSE, in-toto, and SLSA make signed artifacts discoverable and verifiable [35, 52, 57, 58, 61]. Chio uses DSSE and Rekor-compatible ideas but signs action admission, not only artifact provenance. A bilateral Chiodos envelope

says two named kernels independently signed the same invocation predicate after local policy evaluation.

The distinction is precise. A supply-chain provenance statement says how an artifact was built; a Chiodos bilateral statement says how an action was admitted under two policy contexts. The latter may cite the former (for instance when a tool binary or policy bundle needs build provenance), but build provenance alone cannot prove that a buyer admitted a vendor’s receipt under a live treaty continuation.

Selective disclosure work gives the privacy building block. BBS group signatures and Camenisch-Lysyanskaya credential systems underpin modern anonymous credential designs [7, 9]; the CFRG BBS draft standardizes message-vector signing and proof verification [38]. The EU Digital Identity Wallet’s Architecture and Reference Framework mandates SD-JWT VC and ISO mdoc as the first-line attestation formats; ETSI TR 119 476-1 v1.3.1 catalogues SD-JWT, mdoc, BBS+, BBS#, and zk-SNARK overlays as candidate selective-disclosure schemes rather than endorsing any one [24]. Threshold-BBS+ constructions remove the single-issuer-key assumption that the CFRG draft inherits [21, 42]. Chio admits the Ed25519 record into log corpora and uses BBS only at presentation, separating storage from disclosure to avoid the audit-replay-vs-unlinkability tension; the choice is one cell of a larger design space and is fit for 2026 deployment but will need re-evaluation once post-quantum mandates land (salted-hash formats signed under PQC algorithms are ETSI’s recommended migration, not transparent-SNARK substitution).

Governance systems supply the contrast class. Compound and Polkadot OpenGov bind protocol changes to token or stake-weighted processes [13, 46]; Helios provides public auditability for elections [1]; Kleros and Aragon Court move dispute resolution into crypto-economic juries [3, 37]. Optimism’s Token House plus Citizens’ House and ZKsync’s Security Council are the closest production analogs of “polity with electorate plus emergency-amendment path”, with ZKsync’s first-year report recording 4 emergency activations alongside 11 approved proposals [43, 70]. PBFT and Stellar show two assumptions for replicated or federated agreement [10, 39]; Cosmos IBC carries messages across consensus zones with on-chain verification of source-chain commitments [27], and Chio’s bilateral treaty admission is the policy-layer analog where the receiving kernel runs a predicate over a co-signed receipt rather than a chain-rooted Merkle proof. Smart-contract verification (Move Prover, Solidity verifiers) shows that proof-carrying upgrades exist in production for first-order contract properties [5]. Chio’s amendment path adds the obligation absent from all of these: a Lean term witnessing backward refinement before enactment, with emergency overrides surfaced as labeled crisis artifacts rather than ad-hoc Council bypasses.

The governance comparison turns on what a vote can do. Token-weighted systems decide bytes, parameters, or social mandate. Chio’s amendment path requires a proof that the new predicate set refines the old one for admitted history; a vote may choose a candidate, but it cannot by itself enact a constitution that fails the refinement requirement. This separates “on-chain governance” from proof-carrying constitutional change.

Political and legal literature constrains the vocabulary. Ostrom’s polycentric governance work makes local, rule-bound institutions

a serious object of study [44, 45]. Hirschman’s exit and voice frame maps cleanly to Chio citizenship and revocation [32]. Montevideo supplies the declarative statehood vocabulary the paper narrows to digital receipt territory [53]. Srinivasan’s network-state book supplies the motivating but overbroad claim [59]; Walch and De Filippi and Wright warn that blockchain vocabulary outruns legal reality [17, 67]. Chio enforces that warning by binding every state-language claim to a code citation, a named theorem, or an empirical script.

The result sits closer to Ostrom than to a network-state manifesto. A Chio polity is a local rule system with explicit boundaries, monitoring, graduated rejection, conflict artifacts, and exit through revocation or non-admission. The delta from Ostrom is mechanical inspection: predicates, receipts, and proof terms replace deliberative consensus as the medium in which institutional authority is verified.

9 Limitations and Threats to Validity

The proof model is bounded. Table 4 names the cryptographic, library, operational, and legal assumptions used outside it. Lean theorems hold as claims about explicit Chio models, not about the deployed stack.

The remaining limits are:

- **Declarative scope only.** A Chio polity has final authority over admission to its own receipt log; it cannot tax, conscript, expel a physical person, or compel an external court, and Montevideo territoriality is satisfied only in the event-space sense.
- **Theorem inventory is the audited surface.** The curated theorem-inventory entries map proof-tree declarations to protocol properties and Rust symbols; coverage claims rest on inventory links, not on raw declaration counts (which drift with refactoring and may differ across branches).
- **Bounded paper model.** The four new paper-model theorems encode treaty predicate intersection, ladder-floor stability, and amendment as backward refinement; they do not verify the whole Rust implementation.
- **Anchor lane unevenness.** Rekor SET verification is implemented; Rekor Merkle inclusion-proof verification is follow-up code; the Bitcoin-via-OTS lane is verify-real for parsing and commitment linkage and advisory for load-bearing public-witness policy; the Solana memo lane checks root and checkpoint matching, not independent chain finality.
- **Benchmark gaps.** Dispatch, treaty intersection, selective disclosure, and replay corpus have script-backed numbers; receipt signing, receipt verification, and anchor inclusion are withheld until dedicated measurement harnesses exist.
- **TEE and federation surfaces.** The Microsoft Azure Attestation (MAA) TEE verifier is label-only; the TDX, SEV-SNP, and Nitro lanes are implemented; the federation crate still carries legacy stub surfaces alongside the strict Chiodos predicate path.
- **Conflict semantics.** A polity can refuse admission, split logs, or deny treaty entry; it cannot seize another polity’s receipts or treasury unless those assets were already inside its own predicates.

- **Amendment expressiveness.** Backward refinement preserves admitted history; emergency powers, debt restructuring, amnesty, expulsion, and retroactive statutes require explicit crisis artifacts that break refinement and are recorded as such.
- **Legal scope.** Corporate form, data protection, consumer protection, sanctions, insolvency, and venue are human institutional questions a Chio polity produces evidence for, not questions it resolves.
- **No end-user surface.** The paper defines the substrate and evidence path; it does not deliver a constitutional interface, dispute console, treaty negotiation tool, or regulator portal.
- **Key custody at scale.** Per-kernel Ed25519 signing assumes a key-management story (HSM, KMS, TEE-bound key handles) the paper does not specify; rotation, revocation, and compromise recovery for fleets of thousands of kernels are outside the present construction.
- **Partition reconciliation protocol.** Partition-contingency mode is named in the ladder, but the reconnect protocol (how two forked polities exchange reconciliation evidence and resume admission) is not specified; an unsolvable merge collapses partition-contingency to a one-way door.
- **Portability and interop profile.** Receipt schemas are Chio-specific (e.g., `chio.bilateral-cosign-invocation.v1`); a published cross-implementation profile is the prerequisite for any second receiver to admit a Chio receipt without trusting the Chio implementation.
- **Schema evolution across vendor kernels.** Two kernels at different schema versions exchanging receipts currently deny on `unsupported_treaty_scope_schema` without distinguishing the failure from a constitutional denial in operator dashboards; a versioned-predicate compatibility profile and a distinct `schema_mismatch` denial code are required for multi-tenant deployment.
- **Regulator accreditation gap.** No current attestation framework (SOC 2, ISO 27001, PCAOB, sectoral) cites Lean-attested receipt evidence; until a recognized framework maps receipts to existing control language, Chio is an evidence producer with no accredited consumer.
- **Override authority for false negatives.** Fail-closed admission produces a denial receipt rather than a fallback path; the paper does not define which capability authorizes admit-under-protest, what crisis artifact that produces, or how the override is audited.
- **Operational observability.** The receipt log is the audit channel; live observability (metrics, traces, dashboards, denial-rate alerts) requires a non-leaking telemetry path that the present substrate does not specify.
- **Cryptographic-suite migration and PQC trajectory.** The substrate pins Ed25519, BBS, and SHA-256; cross-polity rotation lacks a defined protocol for re-anchoring existing receipt corpora, and the BBS lane in particular will need re-evaluation once post-quantum mandates land. The likely substitution is not a transparent-SNARK upgrade of BBS but a switch to salted-hash formats (SD-JWT VC, ISO mdoc) signed under PQC algorithms, which is the path ETSI TR 119 476-1 and the EUDI ARF actually take.

Table 4: Assumption ledger for the paper’s claims, split into axiomatic foundations and operational discipline.

Assumption	Used for	Residual risk
<i>Foundational (cryptographic / library / legal)</i> Ed25519 and SHA-256 behave as specified.	Receipt signatures, subject digests, canonical hash bindings, Merkle commitments.	Cryptographic break or misuse below the proof model invalidates evidence integrity.
Canonical JSON implementation matches RFC 8785 intent.	Stable receipt identity and DSSE subject digests.	Library defects can create split identities even when Lean predicates are correct.
Legal entities remain outside the kernel.	Regulator-facing interpretation of Chio polities.	Human law can reject, reinterpret, or override any receipt-local governance claim.
<i>Operational discipline (failure modes are policy violations, not breaks)</i> Verifier-owned stores are actually verifier-owned.	Treaty refs, trust bundles, peer pins, revocation epochs, governance receipts, and bilateral cosigners drawn from distinct principals.	Operational shortcuts can reintroduce sender-controlled trust; two-key DSSE under a single actor collapses to one-of-one (kernel-independence attestation is left as a future substrate primitive).
Trust-store-admission predicate is non-amendable.	Closing the constitutional-ratchet attack at the meta layer (the predicate that gates audit-trust).	A meta-stability theorem closing trajectory attacks is future work; the present construction binds the predicate as a non-amendable axiom.
Lean theorem inventory is maintained with code changes.	Mapping proof names to protocol properties and implementation symbols.	Stale inventory can make proof coverage look broader than it is.
Public witness clients verify the statement they claim to verify.	Anchor inclusion and equivocation resistance.	Rekor, OTS, Solana, and EVM lanes have different maturity and finality assumptions.

- **TEE orthogonality.** Receipt cosigning gives offline auditability (canonical-bytes replay) but not online kernel attestation; a TEE-rooted kernel (Intel TDX, AMD SEV-SNP, NVIDIA H100 confidential compute) is a strict strengthening complementary to receipt cosigning, not a competitor.

10 Conclusion

Programmable sovereignty is a substrate property: a polity is a triple (T, C, K) , admission and attenuation and treaty intersection and amendment refinement are decidable kernel operations, Lean theorems anchor the treaty-intersection and amendment-refinement obligations, and a three-vendor buyer-closure is replayable from a fixture-backed generator through the Chio runtime. The construction instantiates condition (a) of the Hartian rule of recognition over cross-organizational tool invocations (the criterion officials apply to identify which rules count), and leaves the practice-of-officials and internal-point-of-view conditions as sociological obligations beyond the present construction. The next concrete result the substrate enables is bilateral admission against a willing co-signing counterparty – an industry consortium, a sectoral self-regulatory organization, or a research-led pilot: verifier-owned trust stores, ladder-floor joins, and governance-receipt requirements are already in place, and the missing input is an external counterparty that publishes a DID and a compatible predicate manifest.

References

- [1] Ben Adida. 2008. Helios: Web-based Open-Audit Voting. In *17th USENIX Security Symposium*. USENIX Association, Berkeley, CA, USA, 335–348. <https://www.usenix.org/conference/17th-usenix-security-symposium/helios-web-based-open-audit-voting>
- [2] Andrew W. Appel. 2015. Verified Software Toolchain. *ACM Transactions on Programming Languages and Systems* 37, 2 (2015), 7:1–7:33. doi:10.1145/2699682
- [3] Aragon Association. 2020. Aragon Court. <https://aragon.org/court>
- [4] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, et al. 2022. Constitutional AI: Harmlessness from AI feedback. arXiv:2212.08073. <https://arxiv.org/abs/2212.08073>
- [5] Massimo Bartoletti, Silvia Crafa, and Enrico Lipparini. 2025. Formal Verification in Solidity and Move: Insights from a Comparative Analysis. In *6th International Workshop on Formal Methods for Blockchains (FMBC 2025)*, OASiCs vol. 129. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 3:1–3:19. doi:10.4230/OASiCs.FMBC.2025.3
- [6] Lukas Berglund, Asa Cooper Stickland, Mikita Balesni, Max Kaufmann, Meg Tong, Tomasz Korbak, Daniel Kokotajlo, and Owain Evans. 2023. Taken out of context: On measuring situational awareness in LLMs. arXiv:2309.00667. <https://arxiv.org/abs/2309.00667>
- [7] Dan Boneh, Xavier Boyen, and Hovav Shacham. 2004. Short Group Signatures. In *Advances in Cryptology – CRYPTO 2004*. Springer, Berlin, Heidelberg, 41–55. doi:10.1007/978-3-540-28628-8_3
- [8] Vitalik Buterin, Sam Wilson, Ansgar Dietrichs, and lightclient. 2024. EIP-7702: Set EOA account code. Ethereum Improvement Proposal (created May 7, 2024; activated in Pectra hard fork, May 2025). <https://eips.ethereum.org/EIPS/eip-7702>
- [9] Jan Camenisch and Anna Lysyanskaya. 2001. An Efficient System for Non-transferable Anonymous Credentials with Optional Anonymity Revocation. In *Advances in Cryptology – EUROCRYPT 2001*. Springer, Berlin, Heidelberg, 93–118. doi:10.1007/3-540-44987-6_7
- [10] Miguel Castro and Barbara Liskov. 1999. Practical Byzantine Fault Tolerance. In *Proceedings of the Third Symposium on Operating Systems Design and Implementation*. USENIX Association, Berkeley, CA, USA, 173–186. <https://www.usenix.org/conference/osdi-99/presentation/practical-byzantine-fault-tolerance>
- [11] Mihai Christodorescu, Earlene Fernandes, Ashish Hooda, Somesh Jha, Johann Rehberger, Kamalika Choudhury, Xiaohan Fu, Khawaja Shams, Guy Amir, Jihye Choi, Sarthak Choudhary, Nils Palumbo, Andrey Labunets, and Nishit V. Pandya. 2025. Systems Security Foundations for Agentic Computing. arXiv:2512.01295. <https://arxiv.org/abs/2512.01295>
- [12] Cloud Security Alliance. 2025. *Agentic AI NIST AI RMF Profile*. Technical Report. Cloud Security Alliance Labs. <https://labs.cloudsecurityalliance.org/agentic/age-ntic-nist-ai-rmf-profile-v1/>
- [13] Compound. 2020. Compound Governance (Governor Bravo). <https://docs.compound.finance/v2/governance/>
- [14] Compound DAO. 2024. Proposal 289 governance attack (July 2024). <https://compound.finance/governance/proposals/289> Carried 682,191 / 633,636 (52% margin); later rescinded by negotiated settlement.
- [15] James Crawford. 2006. *The Creation of States in International Law* (second ed.). Oxford University Press, Oxford, UK.
- [16] Joseph W. Cutler, Craig Disselkoe, Aaron Eline, Shaobo He, Kyle Headley, Michael Hicks, Keshia Hietala, Eleftherios Ioannidis, John Kastner, Anwar Mamat, Darin McAdams, Matt McCutchen, Neha Rungta, Emina Torlak, and Andrew Wells. 2024. Cedar: A New Language for Expressive, Fast, Safe, and Analyzable Authorization. *Proceedings of the ACM on Programming Languages* 8, OOPSLA1 (2024), 123:1–123:30. doi:10.1145/3649835
- [17] Primavera De Filippi and Aaron Wright. 2018. *Blockchain and the Law: The Rule of Code*. Harvard University Press, Cambridge, MA, USA. doi:10.4159/9780674985933
- [18] Markus de Medeiros, Muhammad Naveed, Tancrede Lepoint, Temesghen Kahsai, Tristan Ravitch, Stefan Zetsche, Anjali Joshi, Joseph Tassarotti, Aws Albarghouti, and Jean-Baptiste Tristan. 2025. Verified Foundations for Differential Privacy. *Proceedings of the ACM on Programming Languages* 9, PLDI (2025), 191:1–191:24. doi:10.1145/3729294
- [19] Leonardo de Moura and Sebastian Ullrich. 2021. The Lean 4 Theorem Prover and Programming Language. In *Automated Deduction – CADE 28*. Springer, Cham, Switzerland, 625–635. doi:10.1007/978-3-030-79876-5_37
- [20] Craig Disselkoe, Aaron Eline, Shaobo He, Kyle Headley, Michael Hicks, Keshia Hietala, John Kastner, Anwar Mamat, Matt McCutchen, Neha Rungta, Bhakti Shah, Emina Torlak, and Andrew Wells. 2024. How We Built Cedar: A Verification-Guided Approach. arXiv:2407.01688. <https://arxiv.org/abs/2407.01688>
- [21] Jack Doerner, Yashvanth Kondi, Eysa Lee, and abhi shelat. 2023. Threshold BBS+ Signatures for Distributed Anonymous Credential Issuance. In *2023 IEEE Symposium on Security and Privacy*. IEEE, Los Alamitos, CA, USA, 773–789. doi:10.1109/SP46215.2023.10179470

- [22] Sophia Drossopoulou and James Noble. 2011. Permission and Authority Revisited: Towards a Formalisation. In *Proceedings of the 2011 Workshop on Formal Techniques for Java-like Programs*. ACM, New York, NY, USA, 1:1–1:7. <https://research.google/pubs/permission-and-authority-revisited-towards-a-formalization/>
- [23] Sophia Drossopoulou and James Noble. 2013. The Need for Capability Policies. In *FTFJP 2013 Pre-Proceedings*. ACM, New York, NY, USA, 6:1–6:7. <https://types.cs.washington.edu/ftfjp2013/preprints/a6-Drossopoulou.pdf>
- [24] ETSI. 2025. *Analysis of Selective Disclosure and Zero-Knowledge Proofs Applied to EUDI Wallet*. Technical Report TR 119 476-1 v1.3.1. European Telecommunications Standards Institute. https://www.etsi.org/deliver/etsi_tr/119400_119499/11947601/01.03.01_60/tr_11947601v010301p.pdf
- [25] European Commission. 2025. General-Purpose AI Code of Practice. EU AI Act implementation, in force August 2, 2025. <https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>
- [26] Fuchsia Project. 2020. Capabilities. <https://fuchsia.dev/fuchsia-src/concepts/components/v2/capabilities>
- [27] Christopher Goes. 2020. The Interblockchain Communication Protocol: An overview. arXiv:2006.15918. <https://arxiv.org/abs/2006.15918>
- [28] Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Evan Hubinger, et al. 2024. Alignment faking in large language models. arXiv:2412.14093. <https://arxiv.org/abs/2412.14093>
- [29] Norman Hardy. 1985. KeyKOS Architecture. *ACM SIGOPS Operating Systems Review* 19, 4 (1985), 8–25. doi:10.1145/858336.858337
- [30] H. L. A. Hart. 2012. *The Concept of Law* (third ed.). Oxford University Press, Oxford, UK. First edition 1961.
- [31] Chris Hawblitzel, Jon Howell, Manos Kapritsos, Jacob R. Lorch, Bryan Parno, Michael L. Roberts, Srinath Setty, and Brian Zill. 2015. IronFleet: Proving Practical Distributed Systems Correct. In *Proceedings of the 25th Symposium on Operating Systems Principles*. ACM, New York, NY, USA, 1–17. doi:10.1145/2815400.2815428
- [32] Albert O. Hirschman. 1970. *Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States*. Harvard University Press, Cambridge, MA, USA.
- [33] International Centre for Settlement of Investment Disputes. 2024. Honduras Prospera Inc., St. John’s Bay Development Company LLC, and Prospera Arbitration Center LLC v. Republic of Honduras. <https://www.italaw.com/sites/default/files/case-documents/italaw182539.pdf>
- [34] Gerwin Klein, Kevin Elphinstone, Gernot Heiser, June Andronick, David Cock, Philip Derrin, Dhammika Elkaduwe, Kai Engelhardt, Rafal Kolanski, Michael Norrish, Thomas Sewell, Harvey Tuch, and Simon Winwood. 2009. seL4: Formal Verification of an OS Kernel. In *Proceedings of the 22nd ACM Symposium on Operating Systems Principles*. ACM, New York, NY, USA, 207–220. doi:10.1145/1629575.1629596
- [35] Ben Laurie, Emilia Messeri, and Rob Stradling. 2021. Certificate Transparency Version 2.0. RFC 9162. doi:10.17487/RFC9162
- [36] Xavier Leroy. 2009. Formal Verification of a Realistic Compiler. *Commun. ACM* 52, 7 (2009), 107–115. doi:10.1145/1538788.1538814
- [37] Clement Lesaege, Federico Ast, and William George. 2019. Kleros Short Paper. <https://kleros.io/assets/whitepaper.pdf>
- [38] Tobias Looker, Vasilis Kalos, Andrew Whitehead, and Mike Lodder. 2026. The BBS Signature Scheme. Internet-Draft draft-irtf-cfrg-bbs-signatures-10. <https://datatracker.ietf.org/doc/draft-irtf-cfrg-bbs-signatures/>
- [39] David Mazieres. 2015. *The Stellar Consensus Protocol: A Federated Model for Internet-level Consensus*. Technical Report. Stellar Development Foundation, San Francisco, CA, USA. <https://stellar.org/papers/stellar-consensus-protocol.pdf>
- [40] Mark S. Miller. 2006. *Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control*. Ph.D. Dissertation. Johns Hopkins University. <https://jscholarship.library.jhu.edu/bitstream/handle/1774.2/873/markm-thesis.pdf>
- [41] Toby Murray. 2010. *Analysing the Security Properties of Object-Capability Patterns*. Ph.D. Dissertation. University of Oxford. <https://www.cs.ox.ac.uk/files/3080/thesis-FINAL-15-07-10.pdf>
- [42] Ariel Nof and Vipul Goyal. 2025. Non-interactive Threshold BBS+ from Pseudorandom Correlations. In *Topics in Cryptology – CT-RSA 2025*. Springer, Cham, Switzerland, 184–210. doi:10.1007/978-3-031-88661-4_9
- [43] Optimism Foundation. 2024. Optimism Collective Governance: Token House and Citizens’ House. Governance documentation. <https://community.optimism.io/docs/governance/>
- [44] Elinor Ostrom. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, UK.
- [45] Elinor Ostrom. 2009. Beyond Markets and States: Polycentric Governance of Complex Economic Systems. <https://www.nobelprize.org/prizes/economic-sciences/2009/ostrom/lecture/>
- [46] Polkadot Wiki. 2023. Polkadot OpenGov. <https://wiki.polkadot.com/docs/learn/learn-governance>
- [47] Project Everest. 2017. Project Everest: Verified Secure Implementations for the HTTPS Ecosystem. <https://project-everest.github.io/>
- [48] Joseph Raz. 2009. *The Authority of Law: Essays on Law and Morality* (second ed.). Oxford University Press, Oxford, UK.
- [49] Reuters. 2024. Honduras Top Court Declares Self-Governing ZEDE Zones Unconstitutional. <https://www.investing.com/news/world-news/honduras-top-court-declares-self-governing-zede-zones-unconstitutional-3626537>
- [50] Anders Rundgren, Bret Jordan, and Samuel Erdtman. 2020. JSON Canonicalization Scheme (JCS). RFC 8785. doi:10.17487/RFC8785
- [51] Frederick Schauer. 1991. *Playing by the Rules: A Philosophical Examination of Rule-Based Decision-Making in Law and in Life*. Oxford University Press, Oxford, UK.
- [52] Secure Systems Lab. 2024. DSSE: Dead Simple Signing Envelope (v1.0.2). <https://github.com/secure-systems-lab/dsse>
- [53] Seventh International Conference of American States. 1933. Convention on Rights and Duties of States. https://avalon.law.yale.edu/20th_century/intam03.asp
- [54] Thomas Sewell, Magnus O. Myreen, and Gerwin Klein. 2013. Translation Validation for a Verified OS Kernel. In *Proceedings of the 34th ACM SIGPLAN Conference on Programming Language Design and Implementation*. ACM, New York, NY, USA, 471–482. doi:10.1145/2491956.2462183
- [55] Jonathan S. Shapiro, Jonathan M. Smith, and David J. Farber. 1999. EROS: A Fast Capability System. In *Proceedings of the Seventeenth ACM Symposium on Operating Systems Principles*. ACM, New York, NY, USA, 170–185. doi:10.1145/319151.319163
- [56] Sigstore Project. 2021. Rekord: Software Supply Chain Transparency Log. <https://github.com/sigstore/rekor>
- [57] Sigstore Project. 2021. Sigstore Security Model. <https://docs.sigstore.dev/about/security/>
- [58] SLSA Framework. 2023. Supply-chain Levels for Software Artifacts Specification (v1.0). <https://slsa.dev/spec/latest/>
- [59] Balaji Srinivasan. 2022. *The Network State*. self-published, San Francisco, CA, USA. <https://thenetworkstate.com/>
- [60] Georgios Syros, Anshuman Suri, Jacob Ginesin, Cristina Nita-Rotaru, and Alina Oprea. 2026. SAGA: A Security Architecture for Governing AI Agentic Systems. In *33rd Network and Distributed System Security Symposium (NDSS)*. Internet Society, Reston, VA, USA, 1–19. <https://www.ndss-symposium.org/ndss-paper/saga-a-security-architecture-for-governing-ai-agentic-systems/>
- [61] Santiago Torres-Arias, Hammad Afzali, Trishank Karthik Kuppusamy, Reza Curtmola, and Justin Cappos. 2019. in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes. In *28th USENIX Security Symposium*. USENIX Association, Berkeley, CA, USA, 1393–1410. <https://www.usenix.org/conference/usenixsecurity19/presentation/torres-arias>
- [62] transparency.dev contributors. 2024. Trillian Tessera: A tile-based transparency log. Open-source project, <https://github.com/transparency-dev/tessera>. <https://blog.transparency.dev/introducing-trillian-tessera>
- [63] United States Court of Appeals for the Fifth Circuit. 2024. Van Loon v. Department of the Treasury. <https://law.justia.com/cases/federal/appellate-courts/ca5/23-50669/23-50669-2024-11-26.html>
- [64] U.S. Department of Justice. 2024. Samuel Bankman-Fried Sentenced to 25 Years in Prison. <https://www.justice.gov/usao-sdny/pr/samuel-bankman-fried-sentenced-25-years-prison>
- [65] U.S. Department of the Treasury. 2025. Tornado Cash Delisting. <https://home.treasury.gov/news/press-releases/sb0057>
- [66] W3C. 2022. Decentralized Identifiers (DIDs) v1.0: Core architecture, data model, and representations. W3C Recommendation. <https://www.w3.org/TR/did-core/>
- [67] Angela Walch. 2017. The Path of the Blockchain Lexicon (and the Law). *Review of Banking and Financial Law* 36 (2017), 713–765. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2940335
- [68] Robert N. M. Watson, Jonathan Anderson, Ben Laurie, and Kris Kennaway. 2010. Capsicum: Practical Capabilities for UNIX. In *19th USENIX Security Symposium*. USENIX Association, Berkeley, CA, USA, 29–46. <https://www.usenix.org/conference/usenixsecurity10/capsicum-practical-capabilities-unix>
- [69] Yuhao Wu, Franziska Roesner, Tadayoshi Kohno, Ning Zhang, and Umar Iqbal. 2025. IsolateGPT: An Execution Isolation Architecture for LLM-Based Agentic Systems. In *32nd Network and Distributed System Security Symposium (NDSS)*. Internet Society, Reston, VA, USA, 1–18. <https://www.ndss-symposium.org/ndss-paper/isolategpt-an-execution-isolation-architecture-for-llm-based-agentic-systems/>
- [70] ZKsync Security Council. 2025. ZKsync Security Council year-one report (Aug 2024 – Sep 2025). <https://forum.zknation.io/t/zksync-security-council-report-aug-2024-sept-2025/813>